Relational Schema (ERD), Time-Causal Composite Indexes & Materialized 12-D Feature View

TABLE: high_risk_jurisdictions

PK	country_iso	CHAR(2)
	country_name	VARCHAR(48)
	fatf_status	VARCHAR(24)
	basel_aml_idx	NUMERIC(4, 2)
	sanction_auth	VARCHAR(32)
	effective_date	DATE

Regulatory Sanction Data

1 : N (Country FK)

TABLE: entities_master

PK	entity_id	BIGINT / UUID
	entity_type	VARCHAR(32)
	kyc_risk_tier	SMALLINT (1-5)
FK	country_iso	CHAR(2) -> Jurisdictions
	created_timestamp	TIMESTAMP WITH TZ
	is_pep_flag	BOOLEAN DEFAULT FALSE
	initial_balance	NUMERIC(18, 4)

Primary Entity Vertex Store (N ≈ 100k)

1 : N (Source)

1 : N (Target)

TABLE: streaming_transactions

PK	transaction_id	BIGINT / UUID
FK	source_entity_id	BIGINT -> entities
FK	target_entity_id	BIGINT -> entities
	amount_usd	NUMERIC(18, 4)
	timestamp_utc	TIMESTAMP WITH TZ
	payment_rail	VARCHAR(16) [SWIFT/bKash]
	channel_code	VARCHAR(16) [ATM/API/WIRE]
	conformal_label	SMALLINT [0=Benign, 1=Illicit]

Directed Multigraph Edge Stream (M > 10M)

TIME-CAUSAL COMPOSITE B-TREE INDEX ARCHITECTURE

```
-- Vectorized Zero-Lookahead Causal Indexes
CREATE INDEX idx_source_time
  ON streaming_transactions (source_entity_id, timestamp_utc DESC);
CREATE INDEX idx_target_time
  ON streaming_transactions (target_entity_id, timestamp_utc DESC);
```

- **O(log |E|) Query SLA:** Sub-millisecond 2-hop causal ego-graph extraction in < 0.18 ms
- **Strict Invariant:** t_e <= t_eval strictly enforced (eliminates temporal lookahead leak)
- **Prefix Clustering:** Descending epoch ordering allows zero-copy contiguous range scans

MATERIALIZED VIEW: v_canonical_invariants_12d

Maps Relational SQL Streams -> Universal 12-D PyG Multigraph Tensors

1. Kirchhoff Flow Balance — *Pass-through mule relay (~1.0)*

$$\Phi_{\text{flow}} = \ln(1 + \sum A_{\text{out}} / (\sum A_{\text{in}} + \varepsilon))$$

2. Hawkes Burst Rate — *Micro-burst smurfing spikes (>12.0)*

$$\lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha e^{-\beta(t - t_i)}$$

3. Bi-Directional Taint — *Causal PageRank walk (no leak)*

$$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}} = (1 - \alpha)(\mathbf{I} - \alpha \mathbf{P})^{-1} \mathbf{s}_0$$

4. Ego Multi-Moments — *Higher-order amount/time jitter*

$$[\mu_A, \sigma_A^2, \gamma_A, \kappa_A] \,\|\, [\deg_{\text{in}}, \deg_{\text{out}}, \nu_A, \nu_{\text{io}}] \in \mathbb{R}^8$$

100% Deterministic Feature Basis: Universal Zero-Shot Cross-Ledger Compatibility